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1. Abstract

This project implements large-scale machine learning to the New York City (NYC) Yellow Taxi Trip Records for January 2024. The problem statement is the difficulty in predicting taxi fares and operational productivity due to the various external factors that may have indirect impact on it. Through research study on existing works, this project attempts to build on others' findings and extend the line of inquiry to driver-level productivity predictions using derived metrics relating revenue to time. Linear regression model is used as a baseline interpretation model, followed by random forest regressor which is evaluated for its ability to capture non-linear relationships in taxi fare prediction. A secondary random forest model is trained to assess operational efficiency across trip characteristics. These 3 models are trained on an 80/20 train-test split and evaluated using root mean squared error (RMSE), mean absolute error (MAE), and R^2 . Eventually, the random forest model performs worse than linear regression on the revenue prediction task. On the other hand, the operational efficiency model suggests that shorter, faster trips in high-demand periods yield the greatest productivity for drivers. These findings provide insight with meaningful improvements over classic regression baselines.

2. Introduction

This present work explores using large-scale taxi trip data to support fare prediction and operational efficiency analysis in urban transportation systems. New York City's Yellow Taxi network is one of the most extensively documented transport systems. Despite the availability of this data, accuracy of prediction of fare amount and operational efficiency of individual trips remain to be a challenging problem because fare amounts are affected by variety of factors and there is no single linear relationship that can adequately capture these interactions. This straight forward yet complicated problem motivates the application of machine learning techniques in learning the non-linear patterns from large-scale data. The problem to be addressed is the prediction of taxi trip outcomes, with a primary focus on estimating the total fare amount of a journey and a secondary focus on predicting trip efficiency in terms of revenue generated per minute of travel.

The dataset chosen for this project is the NYC Yellow Taxi Trip Records for January 2024 and it is publicly available on Kaggle with approximately 3 millions data[1]. The aim of the project is to build a dual model approach to accurately predict the total fare amount, *total_amount* for any given trip based on its trip characteristics and the operational efficiency, *rev_per_min* which captures the driver's productivity as revenue per minute. Both objectives are approached using SparkML on the Databricks platform as it enables scalable model training and evaluation over the full dataset. The overall analytical pipeline implementation in this project consists of five stages: dataset cleaning, exploratory data analysis (EDA), feature engineering and selection, model training and evaluation and hyperparameter tuning.

3. Related Work

Due to the public availability of NYC Yellow Taxi trip records and huge data offered, the prediction of taxi fares and outcomes related to the trip have drawn considerable research interests.

One of the studies analysed the trip records using two linear regressions methods implemented on Apache Spark using Ordinary Least Squares (OLS) and the Limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) algorithm for fare prediction [2]. The results of both methods showed similar predictive accuracy with the OLS method performing better in model training speed.

Another study conducted a comparison of multiple machine learning models for taxi fare prediction and resulted in the Random Forest model delivering superior predictive performance compared to simple baselines, with trip distance and passenger count identified as key predictors [3].

More recent work had further extended the line of inquiry to operational performance of the trip records. Zhang (2024) implemented different approach using a multi-objective Long Short-term Memory (LSTM) framework to jointly address fare prediction and operational performance [4]. Rakthao et al. (2025) used data-driven methods to assess taxi profitability and found that metrics relating revenue to time were better in capturing operational efficiency [5].

Compiling these findings together, they form a clear trace of machine learning approach narrative. This project establishes the baseline using classic linear regression method on NYC taxi data for fare prediction, followed by ensemble methods, specifically Random Forest regressor to demonstrate the possible gain through non-linear models. After that, a dual modelling approach is used to predict operational efficiency using efficiency-focused metrics, such as revenue per minute.

4. Research Methodology

4.1 Dataset Selection

The dataset used in this project is the NYC Yellow Taxi Trip Records for January 2024, sourced from Kaggle and loaded into Apache Spark via Databricks platform [1]. The raw dataset contains 19 columns and close to 3 million trip records with numerical, categorical, and temporal attributes including pickup and drop-off timestamps, trip distance, fare amount, tip amount, passenger count, location IDs, vendor identity, rate code, and payment type.

The dataset is selected based on the certain criteria and requirements of the module. First, its huge scale satisfies the module requirement for a dataset with at least 1 million instances. Second, this dataset is a well-established benchmark in the machine learning literature, making it possible to benchmark against the existing published studies. Third, the dataset's temporal attributes support meaningful feature engineering such as, pickup hour, day of week, trip duration, speed and others. These underscore the operational efficiency modelling in the later part of the work. After cleaning the dataset, there are approximately 2.8 million valid records.

4.2 Machine Learning Approach

Linear regression is selected as the interpretable baseline, which is consistent with the approach of Rhouas and El Hami (2025) [2]. Next, the random forest regression is chosen as the primary model. The approach is supported by Huang (2023), who found random forest to outperform linear regression methods on taxi fare prediction [3]. Both models are trained on an 80/20 train-test split and evaluated using RMSE, MAE, and R^2 .

A second model of random forest is trained on the derived efficiency target, *rev_per_min*, extending the analysis from fare prediction to operational efficiency. The use of derived metrics relating revenue to time is inspired by the work of Rakthao et al. (2025) [5]. Finally, a manual grid search tuning is applied to all three models to optimize regularization and tree parameters.

4.3 Data Preprocessing

The NYC Yellow Taxi data was loaded into a reproducible processing pipeline and inspected to establish baseline quality metrics. Initial validation confirmed schema completeness and provided sample records for manual review. The pipeline is designed to produce a clean, documented dataset suitable for descriptive analysis and machine learning.

1. **Remove exact duplicates:** All rows that were identical across every column were dropped to ensure each record represents a single trip event. This prevents double counting and preserves the integrity of aggregate metrics.
2. **Exclude records missing core fields:** Rows lacking essential fare or spatial information were removed. Specifically, records without *fare_amount*, *trip_distance*, *PULocationID*, or *DOLocationID* were excluded because these fields are required for fare calculations and location-based analyses.
3. **Audit and impute passenger count nulls:** We found that many records were missing the number of passengers, which is usually due to system or data entry issues. Since a taxi trip must have at least one passenger, we used the imputation technique to fill in the missing values with 1. This was recorded as a reasonable assumption based on how the data typically behaves.

4. **Outlier review for extreme distances:** Only 35 records exceeded a trip distance of 150 units. Because this is a negligible fraction of the dataset, these records were retained for transparency. The small volume was noted for potential follow up rather than automatic removal.
5. **Apply domain filters to enforce plausibility:** Logical constraints were applied to remove implausible within city trips: `trip_distance > 0`, `fare_amount > 0`, `passenger_count > 0`, and `total_amount < 1000`. These filters eliminate negative or zero values and extremely large totals that are unlikely for urban taxi trips.
6. **Flag fare composition discrepancies:** A derived `total_check` field was computed as the sum of fare components available. There was a large volume of records where `total_amount` and `total_check` diverged, but differences could not be explained with given fare component data points.
7. **Enforce legal passenger capacity:** Records reporting more than five passengers were removed to comply with the legal capacity limit for yellow taxis. This step ensures downstream metrics reflect realistic vehicle occupancy.
8. **Binarize store and forward indicator:** The `store_and_fwd_flag` field was converted from categorical text to a binary numeric indicator. This transformation simplifies model input and enables straightforward aggregation and filtering.
9. **Validate vendor identifiers:** Distinct VendorID values were enumerated and compared against known valid vendor lists. All observed vendor identifiers matched expected values, so no vendor remapping was required.
10. **Encode categorical features for modeling:** Categorical variables such as VendorID, RatecodeID, and `payment_type` were transformed using index encoding followed by one hot vectorization. These representations are compatible with common machine learning algorithms and preserve categorical distinctions.

Every transformation was logged and versioned. Counts of records removed, imputed, and flagged were recorded to quantify data loss and support reproducibility. The pipeline preserves flagged records for reconciliation and includes a short rationale for each rule so that analysts and stakeholders can review assumptions and adjust thresholds as needed.

4.4 Exploratory Data Analysis

This analysis aims to uncover patterns in trip data, including fare trends, passenger behavior, and trip distances, using PySpark on the NYC Yellow Taxi dataset. The key steps and findings are outlined below.

4.4.1 Data Integrity and Preparation

- Trip Distance: The average distance is 3.30 miles, and the maximum recorded distance is 277.40 miles, which still reflects potential outliers that should be removed to ensure reliable analysis. Significant drop-offs in maximum distance highlight the need for effective data cleaning practices.
- Fare Amount: With an average of 18.53 and a maximum of 912.30, the dataset demonstrates clear fare anomalies. These might arise from rare events, erroneous entries, or incomplete data, necessitating further refinement for enhanced accuracy.
- Challenge: lack of geographic coordinates (latitude/longitude) restricted geospatial analysis.

4.4.2 Trip Patterns by Vendor

- Trip Count: Vendor 2 completed a significantly higher number of trips compared to Vendor 1, highlighting a major difference in vendor activity levels.
- Average Fare: Vendor 2 recorded higher average fare amounts than Vendor 1, reflecting differences in trip characteristics or fare structures between the vendors.

4.4.3 Location-Based Insights

- Pickup & Drop-off Hotspots:
 - Top activity clusters included airports and dense urban hubs.
 - Certain zones consistently exhibited higher pickup/drop-off frequencies.

4.4.4 Payment Behavior

- A dominant portion of trips utilized credit cards, reinforcing the prevalence of cashless transactions in urban areas.

4.4.5 Fare and Trip Distance Distribution

- Trip Distance Trends:
 - Short rides dominated (most under 10 miles), with very few exceeding 20 miles.
 - Skewed distribution suggested the dataset focused on urban commuting.
- Fare Patterns:
 - The majority of fares were below \$50, with some high outliers. Fare generally aligned with distance, showing a positive correlation (longer trips equated to higher fares).

4.4.6 Passenger Count Trends

- Single Riders Dominate:
 - Single-passenger trips accounted for the majority of recorded rides.
- Fare vs Passenger Count:
 - Larger groups (7–8 passengers) showed significantly higher average fares, likely reflecting fixed vehicle costs or higher demand for specific services.

4.4.7 Insights

Looking at the data, we noticed that most taxi trips are actually pretty short, usually under 10 miles. This suggests that people are mostly using them for quick city commutes rather than long drives. The fare usually follows the distance, but we did notice some unusually high prices for short trips, which are probably just extra fees or flat rates. We also found that most riders travel alone, and while big groups do not happen often, they do end up paying more on average. Another big trend was that almost everyone uses credit cards now because they are just faster than cash. Even though we did not have exact GPS coordinates, the location of IDs showed that most of the action happens around airports and transit hubs.

4.5 Feature Engineering

The feature engineering pipeline transforms a cleaned PySpark dataframe (*df_transformed*) into a modelling-ready dataset (*df_features*) through four sequential stages: feature construction, exploratory analysis, correlation diagnostics, and importance ranking.

4.5.1 Construction

Seven columns are derived from the raw trip data in a single chained *withColumn* sequence. Three time-related features which include *pickup_hour*, *pickup_dayofweek*, and *pickup_month* are extracted directly from the pickup timestamp. Two trip metrics are calculated: *trip_duration_min* subtracts the pickup from the dropoff timestamp and converts to minutes, while *speed_mph* divides distance by duration in hours, returning zero where duration is zero to avoid division errors. Two categorical encodings complete the set which are *is_weekend* flags Sunday and Saturday trips as 1, and *time_of_day* buckets the pickup hour into morning, afternoon, evening, and night. Three quality filters then remove physically implausible records: trips with zero or negative duration, trips exceeding five hours, and trips where speed exceeds 100 mph.

4.5.2 Exploratory analysis

Average revenue is broken down across every categorical dimension which include time of day, pickup hour, day of week, weekend indicator, month, top pickup zones, and payment type using Spark's *groupBy* and *agg* functions. A 1% sample is drawn into Pandas for histograms and scatter plots. These charts act as a sanity check, confirming the engineered features align with expected demand and pricing patterns before modelling begins.

4.5.3 Correlation

Pearson correlations between each numeric feature and *total_amount* are computed via Spark's distributed *stat.corr()* method and ranked by absolute magnitude. A full 9×9 correlation matrix is then built using *VectorAssembler* and *MLlib's Correlation.corr()*, visualised as a lower-triangle heatmap. This simultaneously identifies the strongest linear predictors and flags inter-feature multicollinearity.

4.5.4–4.5.5 Importance and selection

A Random Forest regressor (100 trees, max depth 8) is trained on a 10% sample, capturing non-linear relationships that Pearson correlation misses. Gini importance scores are extracted and ranked. Trip distance and duration typically dominate, with temporal features contributing modestly and passenger count scoring lowest. Despite this, all eight features are retained in the final list (*final_features*), as the marginal cost of inclusion is negligible. A pairwise multicollinearity scan flags any pair with $|r| > 0.90$ for interpretive awareness, but no features are dropped automatically.

5. Evaluation

5.1 Performance Metrics and Rationale

In this project, regression metrics is selected to align with the business goal of revenue optimization since the task is to predict continuous currency values, in this case, we performed Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-Squared (R^2).

RMSE perceptive, our primary diagnostic for large errors by squaring the residuals. RMSE heavily penalizes significant miscalculations. This is crucial for NYC taxi fleet planning where there is a large error in predicting a high fare airport trip that can disrupt daily revenue targets.

MAE provides a ground level of dollar value of accuracy. The results show that our model is usually between three dollars of the true fare, a highly actionable range for real time driver dispatching.

In terms of R-Squared (R^2), it measures the explanatory power of the features confirms that the engineered variables such as, distance, duration, time and location, account for almost 88% of the volatility in NYC taxi fares.

5.2 Comparison of Baseline and Final Optimized Model

The evaluation measured the optimization from the simple baseline to the final tuned models. Our Baseline was a standard Linear Regression, and Final Model was a hyperparameter tuned Random Forest.

Model Selection	RMSE (USD - \$)	MAE (USD - \$)	R-Squared
Baseline Linear Regression	7.3209	3.5369	0.8867
Tuned Linear Regression	7.3165	3.5344	0.8868
Final Random Forest (Revenue)	7.3847	3.0266	0.8847
Final Random Forest (Operation Efficiency)	0.2638	0.1829	0.8597

An unexpected, however, the critical evaluation insight is that Linear Regression maintained a lower RMSE than the Random Forest, which shows that 2024 taxi fare structure is highly regulated and follows a strong linear progression based on mileage perceptive. On the other hand, the Random Forest achieved a significantly better MAE result (\$3.0266 vs \$ 3.5344), illustrating that the ensemble approach is more reliable for typical city trips where non-liner factors like traffic and zone surcharges are present.

5.3 Case Study: Benchmarking against Contemporary Researches

To fulfill the requirement of the model's output against relevant approaches, we benchmarked our results again three selected studies;

Scenario 1: Profitability & Performance

Compared with Rakthao et al. (2025), their research utilized a data driven simulation to link pricing to operational performance, in that case, their findings align with our evaluation that profitability is not solely a function of distance. By testing our model against high congestion scenario like (low speed/high duration), we confirmed that revenue per minute predictions reflect the same operational performance volatility identified in the Rakthao 2025 simulation.

Scenario 2: Operational Efficiency

From Zhang (2024) research proposed a Multi-Objective LSTM framework, using regional clustering to improve efficiency. In our evaluation indicates that Random Forest Efficiency model which utilizes speed mph and pick_hour as major drivers to achieves a comparable predictive stability for revenue per minute. While Zhang's approach captures spatial temporal clusters, our Spark based collective model provides a lower latency solution that suitable for real time serverless deployment.

Scenario 3: Analytical Robustness

While Sara and Norelislam (2023) study depend on traditional OLS solution, our 2024 model explains for the increased post pandemic traffic density. MAE result \$3.0266 confirms that even though inflation adjusted fare increases between 2023 and 2024, our model maintains a stricter error margin than traditional linear approaches which confirms the ensemble learning works better for modern urban mobility data.

6. Results

6.1 Quantitative Model Comparison

The primary objective of this study was to identify the most effective predictive model for NYC taxi revenue. We compared the baseline performance against tuned version of both Linear Regression and Random Forest models. The table below mentioned the final metric achieved on the hold out test set after manual hyperparameter tuning in summary.

The significant result is that Baseline Linear Regression performed best overall, with the lowest RMSE (\$7.3209) and highest R^2 (0.8867) which highlights that NYC taxi fares are mostly linear, based on the base charges and distance traveled. However, for predicting average trips, Tuned Random Forest performed even better, a lower MAE \$3.0266 compared to \$3.5344 from the baseline model. This means the ensemble model captures real world variation such as traffic and location differences more effectively.

6.2 Visual Insights Feature Relationships

To ratify the quantitative findings, we analyzed multiple key visualizations. These graphs highlight a visual proof of concept for our feature engineering strategy.

Trip Distance vs. Total Amount: The scatter plot in Cell 25 “3.2.16 Scatter plot: Trip distance vs total amount” shows a solid, linear correlation between milage and fare which explains the high performance of Linear Regression baseline.

Trop Duration vs. Total Amount: The scatter plot in Cell 25 “3.2.17 Scatter plot: Trip duration vs total amount” also shows significant vertical fanning which represents trips with similar distances however widely varying duration due to NYC traffic. Our Random Forest model leverages this variance by using trip_duration_min as top three feature that allowing it to predict revenue accurately even in heavy congestion where a purely distance base model would fail.

Numeric Feature Distributions: The histograms in Cell 25 “3.2.10 Histograms for key numeric variables” ensure that most taxi trips are short haul under 5 miles. By describing this distribution, we confirmed that our model is most accurate in the high frequency zone of the business that ensuring peak performance for the majority of the fleet.

6.3 Hyperparameter Tunning/Optimization Insights

Hyperparameter tuning was conducted to optimize model performance while balancing computational efficiency within a distributed Spark framework. A manual grid search strategy was adopted here as opposed to an exhaustive automated process such as cross-validation. This choice was driven to account for both the magnitude of the dataset, which contains around 2.8 million cleaned records, and other factors, such as the size of the Databricks serverless cluster. In that situation, thorough tuning could have proved computationally expensive and even unstable.

6.3.1 Tuning Strategy

For the revenue & efficiency models, all fine-tuning efforts were conducted on the full training split. The iterations were carried out over a defined grid of parameter combinations. Each configuration was evaluated using RMSE, MAE, and R^2 . The optimum parameters were then used to retrain final models to ensure the reported evaluation metrics reflect realistic model performance.

6.3.2 Linear Regression Optimization.

Three hyperparameters were tuned for the Linear Regression model: regParam, elasticNetParam, and maxIter.

Here is how we defined the search space:

- $\text{regParam} \in \{0.01, 0.1, 0.5\}$.
- $\text{elasticNetParam} \in \{0.0, 0.3, 0.5, 1.0\}$.
- $\text{maxIter} \in \{30, 50\}$.

The best results were obtained with $\text{regParam} = 0.01$ and $\text{elasticNetParam} = 1.0$ (measuring a lightly regularised Lasso). This indicates that some regularisation was useful in controlling the complexity of the model. But the response was modest, suggesting that the revenue prediction task already has a strong linear structure supported by trip distance and fare.

6.3.3 Random Forest Optimization.

In the Random Forest model, two of the main hyperparameters were considered in the tuning: number of trees (numTrees) and maximum tree depth (maxDepth).

Hyperparameter	Description	Search Values	Optimal Value
numTrees	Number of trees in the ensemble	{10, 30, 50}	30
maxDepth	Maximum depth of each tree	{5, 7, 10}	10

The best configuration was shown as $\text{numTrees} = 30$ and $\text{maxDepth} = 10$. As more models are generated beyond this threshold, memory consumption increases without corresponding predictive performance benefits.

6.3.4 Tuning Efficiency Models

We tuned a separate Random Forest model for this target variable:

$$\text{rev_per_min} = \frac{\text{total amount}}{\text{trip duration min}}$$

The same tuning grid applied to the revenue-oriented Random Forest was used for the efficiency model. The optimal parameters did not change at $\text{numTrees} = 50$ and $\text{maxDepth} = 10$. The reason for this persistence appears

to be that both of these tasks require a relatively consistent model complexity, even if the significant predictors differ for the two targets.

6.3.5 Lessons of the Tuning process.

What we learned by tuning involved many major takeaways. Initial results, showing that complexity of model was hardly useful, and that simple models, for example, suffice the problems here. Second, the strong results were obtained for linear regression on which it was proved that taxi fare structure is still mostly linear and is strongly associated with distance factor based pricing rules. Third, Random Forest performed with lower MAE, also indicating that non-linear models can better capture regional variations in the average trips like congestion phenomenon and time discrepancy. Finally, the efficiency model demonstrated that speed and time are significant drivers for operational productivity, whereas distance only had relatively minor contributions.

6.4 Different Factors: Revenue Optimization vs. Operation Efficiency

A key takeaway of this analysis is that factors of revenue exist, but they differ from factors that create operational efficiency. Based on results from feature importance, the trip distance predicts total fare revenue the most, but it is considerably less important when predicting revenue per minute. The speed (speed_mph) and pickup_hour, however, generally play the biggest role in the operational process.

This distinction illustrates an interesting operational dimension which might intrigue the driver: rather than taking longer trips, drivers can maintain a higher average speed (or turn trips faster) to maximise earnings. That means in practice, the low congestion and high demand frequency routes do normally better than long haul travel.

The combined use of both models therefore provides complementary insights, enabling a more balanced strategy that considers both total revenue generation and time-based efficiency. This dual perspective can support improved decision-making in fleet management and driver routing strategies.

7. Discussion

Working on this project was a huge learning curve for us, and it really showed us how messy real-world data can be. One of our biggest headaches was dealing with missing values, especially in the passenger count column, which had over 140,000 empty rows. We decided to just fill those with a 1, assuming that every taxi trip has to have at least one person in it. It kept the dataset complete so we could actually run our models, but we know it probably created some bias by ignoring rides with larger groups. If we were to do this again, we would definitely look into better ways to guess those numbers, maybe by looking at the trip distance or the time of day.

Handling outliers was another big hurdle. We found some really unusual entries, like trips over 150 miles or fares that were more than 1,000 dollars. These looked like errors, so we filtered them out to make sure our results were reliable. The downside was that we probably deleted some rare but genuine long distance trips. It was a tough trade-off between having "clean" data and having "all" the data, which is something we really had to wrestle with.

The preprocessing and encoding part were also much more demanding than we expected. We had to turn columns like store and forward flags into binary and use one hot encoding for things like vendor id and payment type. While this was necessary to make the data machine learning ready, it put a lot of strain on the computer. Since we

were using PySpark in a serverless Databricks environment, we really had to focus on optimizing memory and distributing the workload properly so the system would not crash.

When it came to the actual modeling, we chose Random Forest because it is great at picking up complex, non-linear patterns. However, fine tuning the model ended up being one of the most frustrating parts of the project. We had to do a manual grid search for the parameters because the cluster kept hitting memory limits with automated tools. It took a long time to get the tree's depth and count right, and while the fare predictions were strong, it made us realize how much we need better hardware or automated tuning for future projects.

We definitely ran into some limitations we could not avoid. Not having exact GPS coordinates meant we could not do any cool map visualizations; we had to stick to location IDs instead. This worked for big spots like airports, but it was not as detailed as we would have liked. Moreover, by cutting out those extreme outliers earlier, we might have lost some interesting insights into rare trip scenarios.

Looking forward, there is a lot more we could do. We would like to try out gradient boosting or even neural networks to see if they handle the data better. It would also be great to pull in outside information, like weather or traffic data, to see how that affects taxi demand. Eventually, making our PySpark pipeline more scalable would be a huge win so we can handle even bigger datasets without running into the same memory issues.

Overall, this project really highlighted the trade-offs we must make when working with real data. Even though we had to make some tough calls on missing values and outliers, we still got some really solid results. This whole experience taught us exactly how to balance data quality with the scale of the analysis, and it gives us a much better starting point for the next phase.

8. Conclusion

The comparative analysis of the baseline linear regression and the tuned random forest models reveals complementary strengths that can be strategically leveraged to improve both fare prediction accuracy and operational performance. The linear regression model demonstrates superior performance on RMSE and R squared for the revenue task, indicating that it captures the dominant, largely linear relationship between distance-based fare determinants and total fare. This makes it particularly well suited for applications where minimizing large-dollar errors is critical, such as settlement reconciliation, budgeting, and regulatory reporting.

Conversely, the tuned random forest achieves the lowest MAE for revenue predictions and strong performance on the efficiency task, suggesting that it better models the typical trip-level behavior and nonlinear interactions among features such as time of day, pickup/dropoff zones, and driver-specific patterns. Practically, this means the random forest is more effective at reducing average day to day prediction errors that drive operational decisions including dispatching, micro incentives, and routing, while the linear model is preferable when the business must guard against occasional but costly outliers.

To translate these modeling insights into business value, we adopt a hybrid deployment:

1. Use the linear regression for high stakes, long horizon forecasting and revenue protection
2. Deploy the random forest for real time operational decisioning where improving median outcomes yields aggregate efficiency gains.

Both models will benefit from richer, higher quality features like consistent trip time measures, granular temporal indicators, traffic and weather feeds, and driver utilization metrics, along with systematic handling of outliers and fare reconciliation issues identified in the data. Operational interventions that reduce controllable variance in trip times and fares including better dispatch algorithms to cut idle time, targeted driver incentives to smooth supply, proactive vehicle maintenance to reduce breakdowns, and automated fare composition checks to prevent revenue leakage, will not only raise revenue and utilization but also improve model R squared and reduce residual error over time.

Finally, institutionalize continuous monitoring and experimentation including track residual distributions to detect model drift, run A/B tests for pricing and dispatch rules, and retrain models on recent data. By aligning model selection with the specific business objective like minimizing extreme revenue errors versus improving typical operational outcomes and also by iterating on both features and operational processes, taxi vendors can achieve measurable improvements in revenue protection and operational efficiency simultaneously.

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